

Cadence: a patch architecture for continuing intelligence

Learning, memory and action through equilibrium repair

Bernhard Mueller

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Abstract

Cadence investigates a general learning architecture built from bounded observer-like patches that retain state, exchange readings and repair inconsistent predictions. Overlap repair, equilibrium detuning and consequence readback organize a continuing loop: interpret experience, imagine actions, act, and revise the learned relationships from actual outcomes. The goal is reusable intelligence with persistent memory, creative behavior and an evolving functional view of its own proposals. Controlled examples establish narrower capabilities: temporal learning, conditional response protection, action through acquired body models, and exact past-boundary compression under a fixed Gaussian model. Mathematical statements and independently checked experiments identify the assumptions and preserved quantities. Biological mechanisms motivate simplified abstractions; neither brain equivalence nor consciousness follows. Automatic importance, learned specialization, useful recursive self-reflection and reliable long-form creation remain unestablished. Specific control and musical-action tasks serve as tests of the shared architecture, with measured observations kept separate from private imagined futures.

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1 A continuing actor built from local repair

The design question is the smallest persistent state and update rule that preserve useful experience. A capable continuing learner must acquire a body model, retain context and important relationships, imagine consequences, act, and correct itself from what actually happens. Cadence investigates whether these functions can share bounded patch operations instead of requiring a separate mechanism for each task. The individual is the continuing organization of activity, learned relationships, interfaces and body coupling; one scalar energy or a stable output is not a complete description of it.

Observer-Patch Holography (OPH) supplies the organizing constraint: bounded, self-reading patches exchange declared public readings and repair disagreement while protecting observations [25, 26]. Overlap consensus concerns compatible interface values, not identical internal states. Conditional normal-form results distinguish a consistent answer from an answer independent of the initial interior. Neither property establishes that the input evidence is true or that the learned model is useful. Section A.2 gives the exact neural specialization.

Three operations structure the proposal. *Repair* settles local state under held boundaries. *Detuning* changes relationships using local contrasts between equilibria perturbed by real teaching evidence. *Self-readback* makes proposed actions and predicted consequences available to the same continuing actor, whose next observation tests those predictions. Physical readback supplies new evidence; an imagined outcome does not. A preference can guide future action without rewriting the observed past.

These operations have established antecedents. Hopfield networks compute by collective relaxation [10]; equilibrium propagation derives local credit under symmetry and stable-branch assumptions [16, 30]; temporal predictive coding uses transition and readout residuals [18]. Integrated neural architectures and learned body models motivate the whole-actor goal [6, 7]. Cadence’s contribution is an explicit patch/evidence contract, mathematical boundaries on retained state, and controlled tests of which combinations actually support useful action. No superiority over conventional filtering, control or regression follows from sharing this description.

1.1 Scope of the evidence

The base library exposes a nonlinear temporal learner, explicit response constraints, private continuous-input planning and a fixed linear-body actor. These share residual-repair ideas but retain distinct state spaces and hypotheses. The temporal learner accepts time-varying observations, carries only target-free activity after detuning, and can repair action ports under its own acquired dynamics. Both control interfaces hold their body model fixed while separating actual records from hypothetical plans. An audio application must learn or supply its instrument relation. Sources, frozen protocols, controls and receipts identify which implementation supports each result.

The main evidence is presented in Section 3. Appendix A preserves the unique recurrence, record-interference and wiring theorems with their hypotheses. Appendix E retains the source-bound receipt map, including the application evidence that motivated these tests. The public proof package contains 169 audited statements, including the causal-path, quadratic-boundary and protected-path results. Numerical verification is separate from a formal refinement proof of the implementation.

The general behavioral criterion is a learned organization that changes its actual continuation correctly when memory, intention or readback is intervened upon, while preserving unrelated useful experience. In a creative task, it must also construct and evaluate alternatives beyond recalled examples. Musical composition is one demanding instance: heard context, retained intention, imagined alternatives and instrument consequences must influence the same continuing actor.

2 Shared patch operations and persistent state

2.1 Local state, learned relationships and ports

The primary library interface is `TemporalPatchNet`: learned transition/readout maps, carried free activity and explicit temporal boundaries support observed learning and private future repair [20]. `TemporalMemory` adds model-bound response constraints; `plan` changes declared control inputs on a private branch. The temporal model retains A, B, C , free activity, output precision and solver/revision metadata; protection additionally

retains its bases, tolerance and parameter binding. Section 3.1 gives its residual energy. The optimizer and delivery-identifier state below belong to the supported graph specialization.

That graph specialization, **PatchNet**, wraps the nonlinear **Brain** and reciprocal **Learner**. It owns a continuing free neural state, learned synaptic efficacies and biases, and the learner’s optimizer state. Input and output ports name the coordinates through which observations and predictions cross its boundary. A caller supplies their meaning, the initial wiring and the teaching objective. Internal functional specialization is a learning question; declaring a region to be a workspace does not establish that it performs a useful workspace function.

For the graph specialization’s fast state, the local constraint is

$$v_i = \sum_j W_{ij} \text{act}(v_j) + c_i,$$

with the nonlinear activation and repair step specified in Section A. A patch reads its neighbours’ published values and its own boundary drive, then updates its discrepancy. Effective weights, rather than neuron labels or equal neighbouring values, define the relationship being repaired. The software schedules these local dependencies centrally. An energy interpretation requires the effective reciprocal interactions and the activation assumptions of the learning derivation; arbitrary directed connectomes do not acquire that interpretation by sharing the interface.

Its retained computational state has the form

$$S_t = (v_t, \theta_t, o_t, P_t),$$

where v_t is free activity, θ_t the learned relationships, o_t the optimizer state and P_t a bounded optional record of delivered source identifiers. Fixed interface and solver configuration accompany this state. A complete reproducible system also accounts for any external random generator, body state and scheduler that affect subsequent observations. The optional identifier record provides duplicate-delivery bookkeeping; it is not an episodic content store.

Continuous learned weights express graded relationships. The minimal neural core retains activity and parameters without an attached content store or importance classifier. Its empirical obligation is to show which useful memories these states can acquire, preserve and revise; the absence of a separate store does not itself satisfy that obligation.

Operation	State and evidence contract
create	Supplies a reciprocal initial graph and named input/output ports; these are inherited structure, not learned roles.
settle, read	Advances and reads target-free activity without changing learned parameters. Inputs are soft drives and receive recurrent feedback.
observe	Uses externally observed target coordinates and supplied gains; updates parameters only after all required phases converge. A rejected attempt can advance free activity.
imagine	Evolves a sequential private branch; neither live continuation state nor witnessed evidence is changed.
reset	Clears fast activity while retaining parameters, optimizer state and delivery identifiers.
save, load	Preserves continuation state; exact numerical continuation requires the same back-end and precision.

Table 1: The supported graph specialization’s public interaction contract. These operations expose computation and evidence boundaries; they do not guarantee learned competence.

2.2 An explicit temporal overlap

Carrying an initial state cannot preserve its information when complete settlement has one solution independent of that state. The graph specialization’s optional temporal boundary makes the previous free activity $\bar{s}$

an argument of the constraints themselves. With a supplied mask $a_i \in \{0, 1\}$ and strength $\kappa \geq 0$, its energy term is

$$E_{\theta}^{\bar{s}}(s) = E_{\theta}(s) + \frac{\kappa}{2} \sum_i a_i (s_i - \bar{s}_i)^2.$$

The corresponding local recurrent equation adds $\kappa a_i (\bar{s}_i - s_i)$ to the drive. The same $\bar{s}$ is held throughout the free, positive and negative phases of one observed event. It is not advanced at each numerical iteration and is independent of the target’s perturbation strength and evidence gain. The next event uses the preceding free activity; a fresh or reset stream uses a zero boundary. An imagined branch advances only its private previous state.

The default strength is zero, and the default eligible coordinates are the inherited hidden ports. Positive strength is therefore a supplied persistence assumption. It introduces no additional fact store and reuses existing activity, but it changes the model’s equations and must be counted as a structural choice. The unmodified recurrence’s contraction certificate does not automatically certify this changed solver. In a single-minimum landscape, repeated temporal overlap can merely slow forgetting. It does not discover a protected subspace, choose important events or guarantee indefinite retention. The gradient estimate below is conditional on the held boundary; it does not differentiate through preceding events or solve long-delay temporal credit.

An exact scalar example distinguishes persistence from protection. For $a > 0$ and $\kappa \geq 0$, the energy $az^2/2 + \kappa(z - c)^2/2$ has unique minimizer qc , where $q = \kappa/(a + \kappa) \in [0, 1)$. Reusing that minimizer as the next boundary gives $q^n c$, tending to zero. These statements are machine-checked¹. They describe a convex scalar system; they do not prove that every nonlinear neural basin washes out.

A readable trace is a different boundary. The library also provides **Trace**, which a caller connects explicitly to context ports. With movement weighting disabled, it records selected free activity h_t as

$$z_{t+1} = \rho z_t + (1 - \rho)h_t, \quad 0 \leq \rho < 1,$$

and makes z_t readable at declared context ports during the next event. The trace is held fixed across all numerical iterations and nudged phases, then advanced once from the free state. This supplies an explicit fading observation history; carrying a numerical initial state or tethering the immediately preceding activity implements a different operation.

For fixed z_t, h_t , the displayed update is the unique minimizer in u of

$$\frac{\rho}{2}(u - z_t)^2 + \frac{1 - \rho}{2}(u - h_t)^2.$$

Six machine-checked lemmas establish this scalar repair, interval preservation, decay of an initial difference as ρ^n under the same source sequence, and decay to zero under zero source². Here the timescale and source are supplied. If the recurrent source itself depends on the trace, the same-source premise need not hold. These lemmas certify neither a joint energy for the full recurrent system nor its stability, learned importance or permanent memory [R80].

2.3 Readback and detuning as the learning operation

A witnessed target y and an observed-coordinate mask m define a port loss, for example

$$L(s; y, m) = \frac{1}{2} \sum_k m_k (s_k - y_k)^2.$$

Unobserved coordinates do not supply targets. At fixed parameters and fixed temporal boundary, the default centred learner obtains a free state and states under positive and negative perturbations $E_{\theta}^{\bar{s}} \pm \beta L$. It then estimates a parameter derivative from the difference of the corresponding local energy derivatives:

$$g_e(\beta) = \frac{\partial_{\theta_e} E_{\theta}(s^+) - \partial_{\theta_e} E_{\theta}(s^-)}{2\beta}.$$

¹Lean: `repair_unique`, `iterate_closed_form`, `iterate_tends_to_zero`.

²Lean: `unique_minimizer`, `repair_mem_interval`, `same_source_history_difference`, `zero_source_tends_to_zero`.

For the implemented pair interaction, this is an endpoint activity contrast. The current optimizer and parameter constraints determine the ensuing update. The gradient interpretation is imported from equilibrium propagation under its smooth stable-branch, symmetry and convergence assumptions [16, 30]; endpoint locality alone proves none of those assumptions. Finite nudges are biased, branches can differ in a multibasin system, and free-state convergence does not imply convergence of both nudged phases.

The revised interface checks the equation residual of every required phase and refuses a parameter update when a phase reaches its work cap without meeting tolerance. It reports convergence and work rather than treating small successive movement as sufficient evidence of settlement. This is a numerical acceptance contract. Outside the contractive subclass it is neither a bound on distance to a unique equilibrium nor a proof of local stability or correct task predictions.

Phase convergence also does not certify a useful parameter step. The temporal learner’s optional `backtrack` admission keeps the same centered contrast g_β and tests $\theta' = \theta - \eta g_\beta$ by an ordinary free replay from the original observed path boundary. It accepts only a finite, converged prediction satisfying

$$L(\theta') < L(\theta) - \epsilon_{\text{num}}, \quad L(\theta') \leq L(\theta) - 10^{-4} \eta \|g_\beta\|^2,$$

where L is the observed precision-weighted half mean squared error and ϵ_{num} is a declared floating-point floor. At most sixteen successively halved rates are tried. Rejection preserves parameters and their revision count; valid pre-update free activity can advance. Targets score the replay without driving its recurrence, and trial activity is private. This guards improvement on the current observation, not old skills, unseen data or the direction’s exact-gradient interpretation. The fixed-rate default is unchanged. Independent replay of the archived experimental guard checks 961 observed updates, including 576 initial proposals that require a smaller rate. A separate comparison verifies public-interface agreement on seven saved updates; it is not a full training replay through the public implementation [R95].

The distinction between activity repair and learning is essential. A free equilibrium answers the current constraints. A witnessed teaching perturbation changes persistent relationships only through the learning operation. An unobserved read does not supply a fabricated target. The caller’s loss, target and evidence selection therefore determine what counts as a useful correction; a harmful-outcome nudge is supplied valence, not autonomous discovery of importance.

2.4 Memory, imagination and evidence identity

Free activity carries forward to the next decision. Resetting fast activity leaves learned relationships intact, allowing a cold query to test parameter memory separately from a lingering response. Checkpoints preserve parameters, optimizer state, activity and bounded delivery bookkeeping so that restart is not confused with relearning. These contracts make retained state inspectable; task-specific retention must be measured.

Imagined alternatives settle from isolated copies of the free state. They cannot update the live state, learned parameters or witnessed-event bookkeeping. Their predictions can guide an external action, but agreement among simulated branches is not fresh evidence about the world. A branch that predicts a shock and an instrument that reads back a shock have different evidential roles. Learning from a generated target would require an explicitly different, disclosed training protocol.

This boundary is enforced by the branch operation’s lack of writes. The caller supplies observations to the learning interface; the library does not authenticate that a caller-provided target came from a physical measurement. Source identifiers authenticate neither that provenance nor the target’s truth.

An optional source identifier suppresses repeat delivery of the same observed payload while that identifier is retained, and rejects conflicting reuse of that identifier. Finite storage limits this guarantee. Distinct identifiers do not establish independent evidence; repeated reports from one sensor can share a cause. The core does not construct an authenticated causal graph, infer sensor reliability or decide whether a contradiction is noise or a changed world.

2.5 What an equilibrium can remember

Fast memory and parameter memory differ. At fixed drive and parameters, global contraction implies one equilibrium. Consequently complete settlement discards differences carried only by the initial activity. Warm starts save work but do not evade this implication. History can instead affect a persistent parameter,

an explicit boundary value, or a basin choice outside global contraction. A multiple-basin system can preserve a cue under unchanged observations; learning changes its landscape and can destroy that cue while preserving the old cold input-response mapping.

Sufficiency concerns future behaviour. Let h be a history, T a declared deterministic update, and r its required readout. The full future signature is

$$F(h)(u) = r(T^*(h, u)), \quad U(F, i)(u) = F(i :: u), \quad r_F(F) = F(\square),$$

where u ranges over finite input continuations. Histories with identical signatures admit the same shift update and present readout. Every exact memory with state-only updates and exact readout must distinguish histories with different signatures. These statements are machine-checked ³. This is a behavioural quotient, not a finite learned compression algorithm: the signature can be infinite and a callable representation may retain the whole history. A family of N behaviourally distinguishable histories requires at least N distinguishable memory states ⁴. Finite-precision parameters therefore have a capacity limit; counting real coordinates without precision does not remove it. The required task and continuation family determine which distinctions matter. The construction does not reveal arbitrary future importance to a learner [21].

A causal poset supplies dependence, not missing content. Two records with different values can have exactly the same causal order and maximal node identifiers. The corresponding topology-only encoding fails any query that distinguishes those values ⁵. A sufficient frontier must retain the required payloads or boundary summaries as well as causal context. Disjoint replacement writes commute under their update law; conflicting writes on one key do not ⁶. Provenance can therefore support safe delivery and revision, but a DAG alone provides neither storage semantics, learned importance nor confluence.

Normal form is not objective truth. A terminating, confluent repair relation gives a unique normal form from a starting state under its hypotheses. A boundary-determined answer additionally requires all relevant history to be represented in that boundary. Neither condition implies that noisy observations were interpreted correctly. In a two-world construction, the same available reports require latent answers zero and one. Any single estimate z then satisfies

$$z^2 + (z - 1)^2 \geq \frac{1}{2},$$

so at least one squared error is at least $1/4$ ⁷. This is an information bound, not a stochastic noise model. Immediate perfect revision and perfect immunity to an indistinguishable first outlier cannot both be required. Acquiring another informative observation is one possible response to that uncertainty.

Conditional basin retention. Let B be a supplied basin, let an allowed exit from B raise a reference energy by at least $m > 0$, and suppose each time-varying energy E_t differs from that same reference by at most ϵ at every state under consideration. If $2\epsilon < m$, an allowed step that does not increase E_t cannot leave B . Every finite accepted trace therefore stays in B , preserving any readout constant there ⁸. A per-step bound relative only to the preceding energy is insufficient for this conclusion. The theorem assumes the exit margin and allowed moves; it does not certify the numerical neural implementation’s basins, find their margins, or prove retention under arbitrary future learning.

Specialization needs asymmetry and useful consequences. A deterministic permutation-equivariant update preserves an invariant initial state under invariant inputs. Exchangeable owners with identical complete states and exposures cannot differentiate by that update alone ⁹. Random initial conditions, unequal

³Lean: `future_signature_one_step, every_exact_dynamic_encoding_refines_future_signature.`

⁴Lean: `distinguishable_family_capacity_bound.`

⁵Lean: `topology_only_is_insufficient, maximal_ids_only_are_insufficient.`

⁶Lean: `distinct_key_updates_commute, same_key_updates_do_not_commute.`

⁷Lean: `two_target_squared_errors, indistinguishable_noise_or_change_forces_error.`

⁸Lean: `finite_trace_preserves_basin, finite_trace_preserves_readout.`

⁹Lean: `equivariant_rollout_stays_fixed, homogeneous_owner_rollout.`

experience or inherited asymmetry can remove this obstruction. They do not themselves establish useful division of labour. A symmetric task can have asymmetric optima. For a supplied mismatch r and connection price $c > 0$, minimizing $w(r^2 + c)$ over a freely adjustable weight $w \geq 0$ selects $w = 0$ ¹⁰. Retaining useful communication therefore needs a task constraint or consequence beyond cheaper agreement.

3 Architecture and controlled evidence

The temporal experiments extend the patch interface to observed past state, private future state, action and preference ports. Their usefulness is measured by executed consequences and causal controls. Settling accuracy is reported separately from prediction, retention and control accuracy.

3.1 Temporal overlap carries credit only when the constraints carry time

In **TemporalPatchNet**, temporal patch t holds hidden state h_t and public readout y_t . Its transition/readout energy is

$$E_\theta = \frac{1}{2} \sum_t (\|h_t - A\phi(h_{t-1}) - Bu_t\|^2 + \|y_t - C\phi(h_t)\|^2). \quad (1)$$

Adjacent patches overlap on the preceding state; the initial cue u_t is absent after its first event. This residual family is established temporal predictive coding [18]. Detuning the readouts toward a complete observed sequence lets discrepancy propagate across temporal overlaps, so local parameter derivatives can receive credit from later observations.

Merely allocating future patches provides no choice. For deterministic transition and readback maps and a fixed initial state, every finite defect-free path equals the forward recurrence. Lean checks this without a linearity assumption¹¹. The statement concerns vanishing defects, not arbitrary stationary points. Useful planning therefore needs free action ports or additional constraints that make alternative consequences relevant [R83].

In the validated nonlinear sequence comparison, eight hidden coordinates and 80 learned parameters receive 1,536 updates over 16-, 32- and 64-event sequences. A distributed rotation initialization passes the 64/256-event continuation criterion in three development models, but fresh validation passes only one of three initializations on each of three-, four- and six-note motifs. A random orthogonal control passes one of nine. These are separately trained models, with every scheduled outcome retained. Each model receives 114,688 scalar teaching values in complete observed sequences, only during the detuned phases. Independent checks reconstruct derivatives, saved phase witnesses and autonomous outputs; they do not replay every training trajectory. The result supports finite temporal credit and some autonomous sequence acquisition, with neither reliable transfer nor online long-delay learning established [R83].

Coupled components share the same energy. Partitioning $h_t = (c_t, m_t)$ replaces A in Equation (1) by

$$A = \begin{pmatrix} A_{cc} & A_{cm} \\ A_{mc} & A_{mm} \end{pmatrix}; \quad A = M_A \odot W_A,$$

with analogous optional boolean masks for B, C . Both components settle jointly. For batch size N and transition defect e , the derivative for a legal block is $-N^{-1} \sum_{n,t} e_{nt}^a \phi(h_{n,t-1}^b)^T$, multiplied by its mask before the centered contrast and candidate admission. Concatenation is exactly the ordinary recurrence; component names supply no new capacity or specialization. Sequentially training detached components would omit this joint credit. The implemented solver retains dense blocks at total hidden width; zeros do not establish a computational saving [R96].

A bounded comparison gives both a two-group and a dense 64-coordinate model four supplied cyclic rotations of one recorded four-bar phrase, an initial cue, one silent primer and a periodic eight-position clock. A shared CPU cap stops both after 334 updates; all required phases converge and all 668 updates commit. Mean taught sample-identity accuracy is 29.69% for the partitioned model and 76.56% for the dense model, with 4,064 versus 7,104 trainable coefficients. Neither meets the complete recall criterion. Independent replay

¹⁰Lean: `positive_cost_selects_zero`.

¹¹Lean: `defectFree_unique, eq_forward, readback_eq_forward`.

retains both capped outcomes. This single-seed, unequal-parameter comparison does not establish useful specialization [R96].

Initialization affects the available temporal signal. Masking the common orthogonal initializer lowers its spectral radius from 0.95 to 0.812. A separate, untrained diagnostic rescales every surviving recurrent weight by one common factor to restore radius 0.95. Fourth-bar cue-output Jacobian norm rises from 0.000295 to 0.0277, compared with 0.0354 for the dense initializer. This repairs an initialization mismatch, not the trained model’s failure. In general cue sensitivity obeys $S_t = A \text{diag}(\phi'(h_{t-1}))S_{t-1}$: a uniformly contractive path loses a one-time cue, whereas learning can change that path. Spectral radius alone bounds neither nonlinear memory duration nor the masked matrix’s operator norm. The diagnostic establishes larger finite cue sensitivity, not learned timescales or indefinite memory [R96].

3.2 The equilibrium chooses an action using an acquired body relation

The control body obeys $v_{t+1} = v_t + a_t + w_t$ and $p_{t+1} = p_t + v_{t+1}$; the actor reads consecutive positions, not hidden velocity. Inherited coordinates $h_t = (p_t, p_t - p_{t-1})$ expose position and displacement. Six parameters $W = [F \ G]$ learn the next state from real position readings and executed actions. For $E(z) = \|z - Wx\|^2/2$ with quadratic teaching, centered detuning is exactly ordinary squared-regression descent rescaled by $1/(1 - \beta^2)$. At $\beta = 0.1$, all three learned models achieve three-step held-out position MSE below 1.3×10^{-9} after 2,048 transitions. Matched no-history models fail. This is a deliberately transparent acquisition control, not a novel optimizer or learned coordinate system [R84].

With acquired F, G held fixed, future states and actions jointly minimize

$$E^g = \frac{q}{2} \sum_{t=0}^{H-1} \|h_{t+1} - Fh_t - Ga_t\|^2 + \frac{1}{2} \left\| h_H - \begin{pmatrix} g \\ 0 \end{pmatrix} \right\|^2 + \frac{r}{2} \sum_{t=0}^{H-1} a_t^2, \quad q = 1000, \quad r = 0.02. \quad (2)$$

The current observed boundary is fixed; g is a supplied retained preference. Adjacent three-coordinate patches exchange precision matrices and vectors. Only the first settled action is executed, after which the real position replaces the predicted readback and the actor replans. No demonstrated action sequence or external branch selector chooses the command. Soft dynamics constraints can bend imagined physics, so the evaluation measures actual executed cost and prediction defects rather than internal goal satisfaction alone.

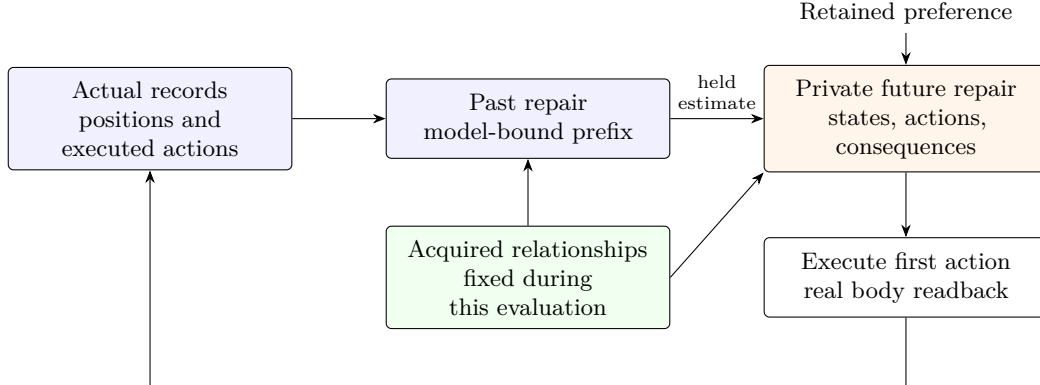


Figure 1: The tested actor’s information flow. Observed records constrain the past; a copied estimate bounds private future repair. A preference changes the proposed action without rewriting the past estimate. Real consequences close the loop. The body relations are learned from separate actual experience and then fixed for this control evaluation.

Controller with acquired body	No disturbance	Hidden disturbance
Three-step feedback repair	0.009675	0.020846
One-step feedback repair	0.034174	0.055940
Initial plan, no later feedback	0.009675	1.234675
Three-step feedback, untrained body	16.38489	17.78535

Table 2: Mean executed cost $(p_3 - g)^2 + v_3^2 + 0.02 \sum_t a_t^2$ on the expanded control grid. Three independently acquired bodies are evaluated at $p_0 = \pm 0.225$, $v_0 = \pm 0.30$, $g = \pm 0.90$ and unannounced first-action disturbance $w_0 \in \{0, \pm 0.35\}$. Lower is better. The disturbance column pools both signs. The retained goal, deadline and physical coordinates are supplied [R84].

All three learned models pass the declared control criteria. Masking new observations reproduces the open-loop recovery failure. The independent audit checks 1,872 episodes across the original and expanded grids and 4,680 plans, including cold restoration, goal reversal and unchanged live state during private planning. Learned feedback actions do not saturate. Forecast error after the disturbed action is 3.01×10^{-10} ; this excludes the unpredictable first disturbance from the forecast-accuracy measure while preserving its full observed error. These controls establish that acquired relationships, future repair and real readback each contribute to action. They do not establish autonomous goal formation or protected memory during further learning [R84].

3.3 Continuous action repair through learned nonlinear dynamics

The same temporal energy supports action search without a separately supplied transition matrix. With parameters and initial boundary fixed, let e_t^+ and e_t^- be transition defects after positive and negative detuning toward a supplied goal. For batch size N , the centered input derivative is

$$g_t = -\frac{(e_t^+ - e_t^-)B}{2\beta N}, \quad L = \frac{1}{2} \text{mean}_{n,t,j} w_j (y_{ntj}^{\text{free}} - y_{ntj}^*)^2,$$

using row-vector defects and positive supplied output weights w_j . Temporal and output normalization enter through the teaching loss. Only declared control ports move; measured ports and the boundary remain fixed. Projected-gradient proposals obey the action bounds, and backtracking accepts one only when an ordinary target-free replay strictly decreases L and meets the Armijo condition. Under smooth stable-branch assumptions the contrast approaches the free-loss derivative as $\beta \rightarrow 0$; finite detuning and solver tolerance introduce error. The returned path is private, with separate reports for projected stationarity and predicted goal attainment. A goal-detuned future is never admitted as a measured outcome [R92].

A fully observed scalar body supplies actual transitions

$$p_{t+1} = 0.85p_t + 0.2 \tanh(2a_t) - 0.05p_t^3, \quad |a_t| \leq 1.$$

Its equation is used only by the environment. A sixteen-coordinate temporal learner receives an initial position, a presence flag and executed actions; subsequent position targets are real readings. Each of three fresh initializations receives 256 updates on batches of eight six-step paths, with 320 learned parameters. Held-out forward mean-square errors are 0.002102, 0.001183 and 0.001648. During control, weights stay fixed, only the first proposed action is executed, and each new measured position initializes another private plan. This fully observed encoding permits a cold hidden boundary; it does not test partially observed memory.

All three models meet the declared forward-prediction and executed-control criteria. Independent arithmetic reconstructs the data, transitions and scores; archived code separately replays all 768 learning updates and 576 plans across 144 episodes. All 288 six-step proposals reach the 32-step cap, despite improving predicted cost; they are not certified stationary optima. One-step control performs similarly on this constant-goal task. These results support learned consequence-guided action and disturbance correction, without establishing anticipation, curiosity, continued adaptation or a recursive self-model [R92].

A sampled-instrument control learns the rendered waveform’s interval RMS amplitude from sample identity and gain, then privately repairs continuous gains toward supplied acoustic goals. One of three scheduled models completes the bounded assessment; the other two have partial or unattempted outcomes, so there is no three-model success claim. With that fixed acquired model, a predetermined gain disturbance is repaired

Controller with learned nonlinear body	No disturbance	Hidden disturbance
Six-step feedback proposal	0.025250	0.025520
One-step feedback proposal	0.025295	0.025565
Zero action	0.180744	0.194487
Random bounded action	0.276985	0.249866

Table 3: Actual eight-step mean squared goal error, averaged across three learned models. Initial position is ± 0.45 , goal is ± 0.35 , and an unannounced position displacement 0 or ± 0.25 occurs before the fifth action. Both feedback horizons use the same learned dynamics; lower is better [R92].

toward an unchanged reference recording: mean squared normalized RMS error falls from 0.09621 to 0.0001712, without training or disclosure of the target gains. Independent checks reproduce the private plan and executed sound. This is restoration under an externally supplied preference. Accurate control of an acoustic goal does not establish that the goal is musically desirable, that the actor learned the listener’s preference, or that it composes original phrases [R95].

3.4 Repair noisy interpretations while protecting measured records

An observation is a fixed record; an interpretation of it can be uncertain. The observer repairs latent position/displacement under actual position records and executed actions. Its Gaussian incoming message summarizes evicted history, with each measurement admitted once. The current mean and covariance are copied before the private goal/action solve. The observer has no goal argument, so a desired future cannot rewrite its factual estimate.

The tested model fixes acquired F, G , sensor variance $R = \sigma^2$, initial mean zero and covariance $100I$, and process covariance $Q = 0.04GG^\top + 10^{-6}I$. These are supplied model assumptions, not inferred reliability or a claim of optimality for the test disturbance distribution. Clean observations are hard position constraints, with a possibly singular posterior covariance. Under noisy readings, repairing the past improves pooled executed cost relative to raw position differences:

Sensor standard deviation	Raw differences	Repaired past
0.02	0.021238	0.021008
0.05	0.042065	0.036596

All 1,296 episodes are retained. At the smaller noise level, cost worsens in six of 27 model/noise/disturbance strata, by at most 0.92%; at the larger level every stratum improves. Independent all-history checks recover means and covariances to 3.91×10^{-13} and 1.22×10^{-15} and reject duplicate-evidence and goal-rewrites-past mutations. This is useful conditional state estimation, not immunity to arbitrary noise or false strong observations [R85].

3.5 A smaller sufficient boundary under a fixed model

For a fixed linear-Gaussian model, the Gaussian prefix is a sufficient boundary for later state inference. A one-record observer is therefore compared with a three-record observer on 27 fresh 64-transition streams. The current mean, covariance and proposed three-step actions agree to 9.05×10^{-13} , 3.39×10^{-14} and 5.83×10^{-12} . The observer’s persistent numeric payload, including a 32-byte model binding, falls from 272 to 208 bytes. Primary past factorizations fall from 5,265 to 1,782; additional prefix marginalizations rise from 1,701 to 1,755. Both kinds of work count. These exogenous streams test equivalence and cost, not another executed-control improvement [R85].

The compressed prefix binds to the values, shapes and dtypes of F, G, Q, R and the fixed prior. Changed models are rejected before the prefix can be silently reinterpreted. An independent sequential Gaussian oracle checks every prefix, with 378 additional dense full-history comparisons. Goal reversals preserve all past estimates and raw records; all 432 attempted operations after model/noise changes reject without modifying the prefix. This is conditional sufficiency at fixed model coefficients. It does not solve how to revise a learned model after discarding the observations that informed it [R85].

A scalar elimination theorem makes the information boundary exact. With fixed coefficients, $p > 0$, $q \geq 0$, and any later objective L that reads only y , define

$$E(x, y) = px^2 - 2bx + q(y - ax - u)^2 + L(y), \quad D = p + qa^2 > 0.$$

Completing the square gives

$$E(x, y) = D \left(x - \frac{b + qa(y - u)}{D} \right)^2 + p'y^2 - 2b'y + c' + L(y),$$

$$p' = \frac{pq}{D}, \quad b' = \frac{q(pu + ab)}{D}, \quad c' = \frac{qpu^2 + 2bqau - b^2}{D}.$$

The reduced expression is an attained lower bound in x . Thus two coefficients preserve later minimization over the retained boundary; the constant also preserves absolute energy. Admitting a new quadratic observation adds its precision and precision-weighted value to these coefficients. Six Lean theorems verify the identities and positivity assumptions¹². The scalar proof and matrix experiment are separate evidence; neither is a theorem that an evolving nonlinear brain can compress all useful history this way [R86].

3.6 Retention and revision are stricter than equilibrium

A local response can be retained by constraining future parameter changes on its supporting activity. For a protected causal path, collect the presynaptic vectors for each shared map: $\tanh h_{t-1}$ for A , u_t for B , and $\tanh h_t$ for C . Let the orthonormal columns of Q_W span the selected vectors for map W . The closest update to a proposed detuning step D_W that leaves these local responses unchanged is

$$D_W^{\text{protected}} = D_W(I - Q_W Q_W^T), \quad D_W^{\text{protected}} Q_W = 0.$$

This is a quadratic parameter-overlap constraint, related to established orthogonal weight modification [38]. Preserving the transition maps on the old path preserves that path by induction; preserving its readback map preserves the emitted values. Three Lean lemmas state this conditional implication without axioms¹³. They assume exact map agreement and do not certify the floating-point SVD, relevance selection or nearby unprotected queries.

The bases compress local response constraints, with no raw phrase or target lookup during inference. The implementation applies ordinary centered detuning and the constrained update atomically. It binds the bases to the admitted parameter history and reports consumed ranks. Protection is selected by the caller and preserves the current learned response, including its existing error. Rank exhaustion can block further learning. Importance, selective fading and justified release of conflicting constraints remain separate questions.

3.7 Musical actions as a controlled example

A bounded real-audio task gives two streams the same eight heard bars and a single initial sign cue, followed by sixteen unobserved bars. The two intended futures are recorded phrases selected by position in pre-existing training splits from two supplied mixes. Future input is zero; targets enter only detuned learning phases. Descriptors use supplied whole-region normalization, so this is not a strict online-amplitude test. A 64-coordinate temporal network has 12,416 learned parameters. Across three initializations, all models meet the declared acquisition, cue-strength, temporal-variation, private-branch and cold-restoration checks. Mean continuation errors are 0.01423, 0.01319 and 0.01329. These are training-phrase realizations with supplied intention, not held-out-source generalization or autonomous goal formation [R88].

An additional third phrase tests further learning while preserving the first response. The protected path consumes ranks 23, 8 and 24 for A , B and C , or 28,328 numeric bytes including tolerance and parameter binding. The teaching cost uses per-output precision

$$w_j = \frac{\text{Var}_t(y_{tj}^{\text{new}}) + 0.0025}{\text{mean}_k[\text{Var}_t(y_{tk}^{\text{new}}) + 0.0025]},$$

¹²Lean: `precision_pos`, `elimination_identity`, `reduced_attained`, `reduced_lower_bound`, `boundary_coefficients`, `admit_observation`.

¹³Lean: `protected_defectFree`, `protected_forward`, `protected_readback`.

computed from the observed new training phrase. These supplied weights add 512 checkpoint bytes; they are task weighting, not learned importance. The evaluation errors and variation gates are unweighted. After 1,536 updates on the new phrase, all three protected models pass the declared acquisition, retention, cue, variation, private-branch and cold-restoration checks. Maximum changes to the old response are 3.92×10^{-14} , 1.02×10^{-14} and 7.44×10^{-15} in independent recurrence checks. New-phrase errors are 0.006226, 0.004330 and 0.003606; variation ratios are 1.464, 1.279 and 1.139, within the fixed $[0.25, 4]$ interval. All three matched unprotected models learn the new phrase but change the old output by 0.992–1.173. Independent checks validate 36 saved phase endpoints and reject eight resealed mutations; they do not replay every intermediate update. This is conditional retention during additional phrase learning, not general lifelong memory [R89].

A separate sampler-grounded test learns actions that produce two recorded one-bar sounds. Sixteen supplied half-beat crops contain the two bars at their native 175 beats per minute. Twenty literal action coordinates specify sample choice, presence, signed offset, gain and duration. A temporal network with 64 hidden coordinates and 5,504 parameters receives a shared activation and an intention sign only at the first of eight events; later input is zero. Source-derived action sequences supply teaching targets, and known-action playback checks the sampler before training.

After 1,536 updates, all three development seeds meet the declared recall, cue, timing, restoration and private-imagination checks. All 48 sample choices across the two intentions and three seeds are correct; reversing the cue selects the other taught sequence. Independent recurrence, saved-phase and exact audio replay checks bind the learned action path to its sound. The listening files use the prescribed first seed and repeat one generated bar four times. Timing within each crop is inherited from the recording. This establishes conditional sample-sequence recall and sound realization; it does not establish original composition, auditory-feedback learning, fresh-seed generalization or retention after learning another task. A learned instrument relation, useful higher-level readback and coherent long-form development are separate requirements [R90].

A four-bar repertoire uses 32 recorded slices, a two-coordinate initial cue and a 64-coordinate temporal learner with 6,592 parameters, jointly trained for 1,536 updates. All three models meet the declared recall criteria, with 96 of 96 taught sample choices correct. Three untaught cue combinations produce mixed slice orders; all are rendered from the prescribed first seed without ranking. Independent checks reproduce their PCM and MP3 files exactly. The task has no absolute mean-square-error gate. It tests joint repertoire acquisition and cue interpolation, not sequential retention or deliberate composition [R93].

Response protection can leave usable but poorly conditioned directions. In a three-program sampled-action task, protection consumes ranks 14, 2 and 16 for A , B and C , or 15,432 numeric bytes, leaving 4,160 parameter directions available. The fully constrained input map fixes the cold first activity q . If Q spans protected readout activity and $r = (I - QQ^T)q$, every correction with $\Delta CQ = 0$ and $(C + \Delta C)q = y^*$ satisfies

$$\|\Delta C\|_F \geq \frac{\|y^* - Cq\|_2}{\|r\|_2}.$$

Residual activity norms of 0.0245–0.0322 require readout changes of norm 21.9–28.8 for exact first-output correction. This bound concerns conditioning, not rank exhaustion or whole-program impossibility [R91].

A readout metric addresses this numerical constraint without teaching from protected targets. Let X contain free presynaptic activities, $S = X^T X / (NT)$, $P = I - QQ^T$ and g_C be the same centered detuning contrast. With supplied damping $\mu > 0$, the update

$$\Delta C = -\eta g_C P(PSP + \mu I)^{-1} P$$

uniquely minimizes $\langle g_C, D \rangle + \text{tr}[D(S + \mu I)D^T] / (2\eta)$ subject to $DQ = 0$. The conditioning matrix uses free activity and protected bases; teaching enters through the same contrast; A and B retain ordinary projected detuning. This is a patch-level dense solve, not an independent per-synapse operation. The metric uses S , whereas the fixed-feature half-mean-loss Hessian is S/O for O output coordinates. Damping and scaling are declared choices, not learned importance [R94].

The matched test uses the same acquired two-bar models and 1,536 updates on a supplied third sample program in both arms, with $\beta = 0.001$, $\eta = 1$ and $\mu = 10^{-4}$ for the metric. All three metric models preserve both old responses within 4.74×10^{-14} and fit the complete new action path below 3.48×10^{-23} mean squared error. All three ordinary projected models preserve the old responses but have new errors 0.008546, 0.011507 and 0.011048. The new-path thresholds are 0.001 and one quarter of its initial error. Thus the metric passes all three joint retention/acquisition criteria; ordinary projected detuning passes none. Every model chooses all

sixteen old and eight new sample identities correctly, illustrating why category accuracy alone is insufficient. The prescribed first seed preserves both old rendered WAV files byte for byte. Independent checks replay every one of the 9,216 updates, all 27,648 phase records and twelve PCM files, with separate constrained-optimum and finite-difference witnesses [R94].

This result has a narrow numerical scope. The metric adds one 64×64 solve per update and no persistent optimizer state; readout changes have norm 32.88–42.20, compared with 1.62–1.68 for ordinary updates. Off-path noise sensitivity is unmeasured. Positive curvature at both detuned endpoints does not establish a common stable branch: at a fixed checkpoint, $\beta = 0.1$ produces an ascent direction despite positive endpoint curvature. At the final near-zero losses, relative gradient errors are dominated by solver tolerance. The matched test has no parameter backtracking; one seed was used in development and two extend that fixed choice. Protection and the new program are supplied. The result establishes conditional acquisition with retention, not automatic importance, selective forgetting, unlimited capacity or original composition [R94].

4 Biological abstractions and functional self-readback

Animal and human brains motivate distributed recurrent interpretation, different memory timescales, learned action consequences and specialization through unequal experience. These are functional design constraints, not a claim that Cadence reproduces the brain’s anatomy or dynamics. Persistent-activity theories motivate carried context [9, 34]; complementary-learning-systems theories motivate separating fast experience from slower shared structure [15, 17]; predictive-coding models motivate local discrepancy signals [5, 29]. Each abstraction must earn its inclusion by retaining information or improving a measured decision.

Functional self-readback requires a causal role. Feeding $y = C\phi(h)$ back into $h' = A\phi(h) + Dy$ is equivalent to $h' = (A + DC)\phi(h)$; it does not create additional evidence or memory. An explicit matrix control verifies this reparameterization [R97]. The control experiment instead compares predicted consequences with new physical readings and demonstrates that masking those readings impairs recovery. An evolving higher-level view is a design goal: patches could read summaries of proposed actions, uncertainty or remembered constraints and revise those summaries from outcomes. Its usefulness requires a matched causal intervention: holding observations, state budget and compute fixed, remove or shuffle that view and measure the behavioral change. Recursively stacking interfaces alone proves no such function. This paper makes no consciousness claim.

Specialization must also be useful. Exchangeable patches can receive unequal experience or break initial symmetry, but unequal weights or a named region do not establish division of labor. A learned division should improve behavior under causal lesion and transfer tests while preserving unrelated records during further learning. The fixed-model boundary and supplied-address retention results state concrete cases where information can be separated; discovering the required separation in music is a different problem.

5 Limits and the generalization test

The control results cover acquired linear dynamics with inherited coordinates and a fully observed scalar nonlinear body, each with supplied goals and acquired dynamics frozen during execution. Longer planning helps the linear task but supplies no established anticipation benefit on the scalar nonlinear task. Gaussian compression is exact only relative to its fixed linear model. The real-audio results use complete teaching sequences, three initializations and selected training phrases; held-out musical generalization is unestablished. Exact response protection is conditional on selected local constraints and usable plastic directions; the sampled-action comparison shows that a supplied readout metric can address poor conditioning despite unconsumed rank. Its large coefficients and fixed damping require further robustness tests. These are not general lifelong-memory guarantees.

Generalization requires useful behavior across new circumstances, retained experience and changed body consequences. For composition, the decisive test is joint: a brief intention must survive a shared heard prefix; changed intention must change the continuation; altered instrument consequences must change the next plan; imagined branches must leave witnessed state untouched; and learning new material must preserve and, when appropriate, revise old material. Context length, autonomous variation and low one-step error are separate measurements. Short intention-conditioned phrase acquisition and explicit response protection establish parts of this standard. The complete learned instrument, self-readback and long-form composition

loop is unestablished. Training on the available jungle corpus and creating original whole tracks is an intended application, not a result demonstrated by these one-bar controls. The same intervention standard must hold in other sensory and action domains; adding modalities supplies no universal scaling guarantee.

Total cost includes all free and detuned phases, failed numerical attempts, matrix factorizations, communication, model acquisition, inference and any retained evidence. Local parameter derivatives do not imply low wall-clock cost. A conventional regression, filter, recurrence or controller is a necessary comparison when it solves the same stated problem. Learned developmental wiring, importance selection and robust transfer across tasks require evidence beyond the fixed small worlds measured here.

6 Conclusion

Cadence seeks a small set of reusable operations for continuing intelligence: repair interpretations, learn relationships through detuning, and compare proposed consequences with actual readback. Controlled bodies and musical actions establish useful instances of these operations. Explicit memory constraints and sufficient boundaries state what can be preserved; a matched readout-metric comparison demonstrates further acquisition while retaining two constrained responses. Creative behavior and evolving functional self-reflection remain architectural goals requiring causal tests. Automatic importance, learned specialization, robust transfer and original whole-track composition are not established. Progress is measured by those joint capabilities and their total cost, not by equilibrium alone.

References

- [1] David H. Ackley, Geoffrey E. Hinton, and Terrence J. Sejnowski. A learning algorithm for Boltzmann machines. *Cognitive Science*, 9(1):147–169, 1985. doi: 10.1207/s15516709cog0901_7.
- [2] Jimmy Ba, Geoffrey E. Hinton, Volodymyr Mnih, Joel Z. Leibo, and Catalin Ionescu. Using fast weights to attend to the recent past. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL <https://arxiv.org/abs/1610.06258>.
- [3] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://arxiv.org/abs/1909.01377>.
- [4] Stefan Banach. Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3:133–181, 1922.
- [5] Andre M. Bastos, W. Martin Usrey, Rick A. Adams, George R. Mangun, Pascal Fries, and Karl J. Friston. Canonical microcircuits for predictive coding. *Neuron*, 76(4):695–711, 2012. doi: 10.1016/j.neuron.2012.10.038.
- [6] Josh Bongard, Victor Zykov, and Hod Lipson. Resilient machines through continuous self-modeling. *Science*, 314(5802):1118–1121, 2006. doi: 10.1126/science.1133687.
- [7] Chris Eliasmith. *How to Build a Brain: A Neural Architecture for Biological Cognition*. Oxford University Press, 2013. ISBN 9780199794546. doi: 10.1093/acprof:oso/9780199794546.001.0001. URL <https://academic.oup.com/book/6263>.
- [8] Nicolas Frémaux and Wulfram Gerstner. Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9:85, 2016. doi: 10.3389/fncir.2015.00085.
- [9] Patricia S. Goldman-Rakic. Cellular basis of working memory. *Neuron*, 14(3):477–485, 1995.
- [10] J. J. Hopfield. Neurons with graded response have collective computational properties like those of two-state neurons. *Proceedings of the National Academy of Sciences*, 81(10):3088–3092, 1984. doi: 10.1073/pnas.81.10.3088.
- [11] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: fast autoregressive transformers with linear attention. In *International Conference on Machine Learning*, 2020. URL <https://arxiv.org/abs/2006.16236>.
- [12] Teuvo Kohonen. Correlation matrix memories. *IEEE Transactions on Computers*, C-21(4):353–359, 1972. doi: 10.1109/TC.1972.5008975.
- [13] Dmitry Krotov. A new frontier for Hopfield networks. *Nature Reviews Physics*, 5:366–367, 2023. doi: 10.1038/s42254-023-00595-y.
- [14] Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL <https://arxiv.org/abs/1606.01164>.

- [15] Dharshan Kumaran, Demis Hassabis, and James L. McClelland. What learning systems do intelligent agents need? complementary learning systems theory updated. *Trends in Cognitive Sciences*, 20(7):512–534, 2016. doi: 10.1016/j.tics.2016.05.004.
- [16] Axel Laborieux, Maxence Ernout, Benjamin Scellier, Yoshua Bengio, Julie Grollier, and Damien Querlioz. Scaling equilibrium propagation to deep ConvNets by drastically reducing its gradient estimator bias. *Frontiers in Neuroscience*, 15: 633674, 2021. doi: 10.3389/fnins.2021.633674.
- [17] James L. McClelland, Bruce L. McNaughton, and Randall C. O’Reilly. Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3):419–457, 1995. doi: 10.1037/0033-295X.102.3.419.
- [18] Beren Millidge, Mufeng Tang, Mahyar Osanlouy, Nicol S. Harper, and Rafal Bogacz. Predictive coding networks for temporal prediction. *PLOS Computational Biology*, 20(4):e1011183, 2024. doi: 10.1371/journal.pcbi.1011183.
- [19] Javier R. Movellan. Contrastive Hebbian learning in the continuous Hopfield model. *Connectionist Models: Proceedings of the 1990 Summer School*, pages 10–17, 1991. doi: 10.1016/B978-1-4832-1448-1.50007-X.
- [20] Bernhard Mueller. Cadence: machine learning by settling nets with owner-local learning, 2026. URL <https://github.com/muellerberndt/cadence>. Python library, MIT license; measured implementation versions and source hashes are identified in the accompanying evidence package.
- [21] Bernhard Mueller. Cadence memory: representation boundaries and local-repair diagnostics, 2026. Executable experiments, source-bound receipts, independent verifiers and Lean 4 proofs in the accompanying evidence package.
- [22] Bernhard Mueller. Verified projection-event calculus in Lean 4: Bundled arbitrary-partition pinching, Lüders retractions, and CHSH interoperability, 2026. URL <https://floatingpragma.io/oph/papers/machine-checked-finite-event-algebras/>.
- [23] Bernhard Mueller. Observer patch nets: local repair, retained state and the console evidence, 2026. URL <https://github.com/muellerberndt/reverse-engineering-reality>. Observer patch holography corpus.
- [24] Bernhard Mueller. Thinking as patch-net fixed-point search, 2026. URL <https://github.com/muellerberndt/reverse-engineering-reality>. Observer patch holography corpus.
- [25] Bernhard Mueller, Jinwook Kim, David Matscheko, and Jonathan Hill. Observation-determined normal forms: Stability, obstructions, and refinement in constraint and rewrite systems, 2026. URL <https://floatingpragma.io/oph/papers/observation-determined-normal-forms/>. Observer patch holography corpus, machine-checked in Lean 4 with Mathlib.
- [26] Bernhard Mueller, Kai Xue, Jinwook Kim, Arnav Anirudha Kale, David Matscheko, and Jonathan Hill. Reality as a consensus protocol: The fixed-point computation that implements physics, 2026. URL <https://floatingpragma.io/oph/papers/reality-as-consensus-protocol/>.
- [27] Randall C. O’Reilly. Biologically plausible error-driven learning using local activation differences: the generalized recirculation algorithm. *Neural Computation*, 8(5):895–938, 1996. doi: 10.1162/neco.1996.8.5.895.
- [28] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2008.02217>.
- [29] Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2:79–87, 1999. doi: 10.1038/4580.
- [30] Benjamin Scellier and Yoshua Bengio. Equilibrium propagation: bridging the gap between energy-based models and back-propagation. *Frontiers in Computational Neuroscience*, 11:24, 2017. doi: 10.3389/fncom.2017.00024.
- [31] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning*, 2021. URL <https://arxiv.org/abs/2102.11174>.
- [32] Wolfram Schultz, Peter Dayan, and P. Read Montague. A neural substrate of prediction and reward. *Science*, 275(5306): 1593–1599, 1997. doi: 10.1126/science.275.5306.1593.
- [33] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.
- [34] Xiao-Jing Wang. Synaptic reverberation underlying mnemonic persistent activity. *Trends in Neurosciences*, 24(8):455–463, 2001.
- [35] Bernard Widrow and Marcian E. Hoff. Adaptive switching circuits. *IRE WESCON Convention Record*, 4:96–104, 1960.
- [36] Xiaohui Xie and H. Sebastian Seung. Equivalence of backpropagation and contrastive Hebbian learning in a layered network. *Neural Computation*, 15(2):441–454, 2003. doi: 10.1162/089976603762552988.
- [37] Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL <https://arxiv.org/abs/2406.06484>.
- [38] Guanxiong Zeng, Yang Chen, Bo Cui, and Shan Yu. Continual learning of context-dependent processing in neural networks. *Nature Machine Intelligence*, 1:364–372, 2019. doi: 10.1038/s42256-019-0080-x.

A Certified recurrence and local credit

These statements concern the declared recurrent subclass; a prototype must meet their hypotheses before using their guarantees.

A.1 Neurons, synapses, settling

A net has n neurons. Neuron i holds a potential $v_i \in \mathbb{R}$ and publishes an activation $s_i = \text{act}(v_i)$. Synapses carry effective weights W_{ij} from neuron j to neuron i ; in the library the effective weight is the product of a gain, the number of contacts, a learned efficacy and a presynaptic factor, and the paper writes the product as W_{ij} . A stimulus c_i is an additive drive, with the bias absorbed into it. One settling step is

$$v_i \leftarrow (1 - dt) v_i + dt \left(\sum_j W_{ij} \text{act}(v_j) + c_i \right), \quad 0 < dt \leq 1. \quad (3)$$

The activation is a rectified, re-based logistic. With slope λ , threshold θ , leak $\ell \in [0, 1]$ and $\sigma(x) = 1/(1 + e^{-x})$,

$$\rho = \sigma(-\lambda\theta), \quad r(v) = \sigma(\lambda(v - \theta)) - \rho, \quad \text{act}(v) = \frac{\max(r(v), 0)}{1 - \rho} + \ell \frac{\min(r(v), 0)}{\rho}. \quad (4)$$

Rest publishes exactly zero, the activation rises to one, and a neuron below rest publishes a small negative value so that the learning rule can move it. The learning model uses $\lambda = 1$, $\theta = 0$, $\ell = 1/10$. A settling run stops when the largest activation movement in a step falls below a tolerance or when a step cap is reached, and the library reports the residual of equation (3) at the final state. Adaptation, a slow variable subtracted from the drive, produces rhythmic dynamics; the theorems below are stated without it.

A declared neural graph can include sensory, internal and motor coordinates. Joint settlement can complete a partial reading when its learned constraints support that completion; region names alone assign no function.

Intuition: each neuron checks its own account. Imagine an arm whose visual input has moved while its motor and body predictions describe the preceding position. Each neuron compares its held potential with the drive its neighbours and senses supply. Equation (3) moves the potential partway toward that drive; the published activation changes what neighbouring neurons receive, and they make the same comparison. The resulting computation is a feedback process with a measurable discrepancy at every step. In the observer-patch description, each bounded patch owns state, communicates through declared ports, reads back its consequences and retains records. A settling certificate and the public evidence bundle make that self-reading process checkable. The equilibrium reconciles the equations the model implements; whether those equations describe the arm well is a separate empirical question.

A.2 From observable events to a global normal form

OPH distinguishes internal descriptions from shared readings [24–26]. A visual patch may locate an object in camera coordinates and a motor patch in body coordinates. Agreement concerns the position after translation to a shared interface. A change of internal coordinates and corresponding interface map is a gauge change when it preserves every declared public reading. Gauge-invariant repair corrects disagreement in those readings.

For local states x_i and readout maps $\pi_{i,e}$ into a common interface on an overlap $e = \{i, j\}$, agreement means

$$\pi_{i,e}(x_i) = \pi_{j,e}(x_j).$$

A repair preserves the observations held fixed. OPH’s local-repair criterion requires every allowed fixed exterior to admit a consistent interior. A *normal form* has no repair left to make. For a terminating discrete relation on the observable quotient, local confluence gives one normal form from each starting state. Completeness, meaning that normal states coincide with consistent states, makes it a global solution. Local compatibility alone need not imply global consistency. For different interiors with the same protected observations to yield the same answer, the compatible consistent states must also form one equivalence class [25, 26].

What counts as an observable event? For a finite classical record, events are sets of possible readings, closed under union, intersection and complement. OPH’s verified projection-event calculus supplies conditional expectations that discard internal distinctions while preserving the probabilities of the declared partition events [22]. Cadence uses classical activation ports and writable records; the quantum projection and conditioning machinery is mathematical background, without a claimed implementation in this neural model.

The neural specialization. Hold the weights fixed, protect $B(c, v) = c$, and define consistency by $v = W \text{ act}(v) + c$. Under Cadence’s contraction condition, every initial state at a fixed input approaches the same equilibrium. The residual and stimulus bounds quantify accuracy and sensitivity. This realizes an observation-determined equilibrium: local equations determine one global answer with a checkable error bound. Numerical convergence replaces finite exact termination; the synchronous proof does not establish arbitrary asynchronous or transactional repair guarantees. Learning determines whether the answer is useful.

A.3 The certificate: how much work a decision needs

Write $F_c(v)$ for the right-hand side of (3) at stimulus c , $\|\cdot\|_\infty$ for the sup norm on $\mathbb{R}^n$, $\rho_W = \max_i \sum_j |W_{ij}|$ for the largest absolute incoming weight sum (the row mass), and L for a Lipschitz constant of act .

Theorem A.1 (settling certificate). *If $0 < dt \leq 1$ and $L\rho_W < 1$, then F_c is a contraction of $(\mathbb{R}^n, \|\cdot\|_\infty)$ with constant $q = 1 - dt(1 - L\rho_W) < 1$.*¹⁴

Corollary A.2 (unique equilibrium). *Under the hypotheses of Theorem A.1 the net has exactly one equilibrium v_c^* , and every fixed point of F_c equals it; with zero stimulus and $\text{act}(0) = 0$ that equilibrium is rest.*¹⁵

Theorem A.3 (a priori and a posteriori certificates). *Under the hypotheses of Theorem A.1, for every start v and every k ,*

$$\|F_c^k(v) - v_c^*\|_\infty \leq \frac{q^k}{1 - q} \|v - F_c(v)\|_\infty, \quad \|F_c^k(v) - v_c^*\|_\infty \leq \frac{\|F_c^k(v) - F_c^{k+1}(v)\|_\infty}{1 - q}.$$

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A residual below ϵ at the returned state certifies a potential error of at most $\epsilon/(1 - q)$, and an activation error of at most L times that. This is an error bound relative to the model’s equilibrium, not a certificate of prediction accuracy. Banach’s theorem is the whole proof [4]; the activation-specific condition is given in Theorem A.6. A numerical control checks the bound on 57,600 cases [R8].

Theorem A.4 (the equilibrium is Lipschitz in the stimulus, and warm starts pay). *Let F_c and $F_{c'}$ satisfy Theorem A.1 with the same constant q and let $\delta = \|c - c'\|_\infty$. Then*

$$\|v_c^* - v_{c'}^*\|_\infty \leq \frac{dt \delta}{1 - q}, \quad \|F_{c'}^k(v_c^*) - v_{c'}^*\|_\infty \leq \frac{dt \delta q^k}{1 - q} \quad \text{for every } k.$$

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This is the theorem behind adaptive work. Starting from the old equilibrium, the number of steps that reaches accuracy ϵ after an input change δ is at most the nonnegative ceiling of $\log(dt \delta / ((1 - q)\epsilon)) / \log(1/q)$: logarithmic in the size of the change, zero when the change is zero. The bound is sufficient rather than a prediction of semantic difficulty, and checking a returned state has its own cost.

¹⁴Lean: `settle_contracting`.

¹⁵Lean: `equilibrium_unique, silence`.

¹⁶Lean: `apriori_bound, aposteriori_bound`.

¹⁷Lean: `equilibrium_lipschitz_stimulus, warm_start_bound`.

Intuition: reuse the answer, then bound the repair. Contraction says that two possible internal states move closer under the same input: after one step their largest difference is at most q times its previous value. With $q = 1/2$, every step halves that uncertainty. If the residual at a returned state is 10^{-3} , the remaining potential error is at most $2 \cdot 10^{-3}$; the factor $1/(1-q)$ sums all the smaller corrections that can follow. When a sensor changes slightly, the preceding equilibrium is a useful starting point because Theorem A.4 bounds how far the required answer can move. This can save repeated work on a slowly changing stream. It gives little advantage after a large change, and convergence can be expensive when q is close to one. A unique equilibrium also erases differences in starting state: lasting history must enter through records, traces or a changed drive, rather than through the warm start alone.

Theorem A.5 (bounded damage). *Let $|\text{act}| \leq A$, let S be a set of neurons, let W^S be W with the columns in S set to zero, and let $\mu_S = \max_i \sum_{j \in S} |W_{ij}|$. If F_c satisfies Theorem A.1, then the lesioned map satisfies it with the same constant, and $\|v^* - v^{*,S}\|_\infty \leq dt \mu_S A / (1 - q)$.¹⁸*

Theorem A.6 (the library’s neuron). *For $\lambda \geq 0$ and $\ell \geq 0$, the activation (4) is Lipschitz with constant $\frac{\lambda}{4} \max(\frac{1}{1-\rho}, \frac{\ell}{\rho})$, satisfies $|\text{act}| \leq \max(1, \ell)$, and publishes exactly zero at rest. Under the learning model the constant is $1/2$ and the bound is 1, so every net whose row mass is below 2 settles by a certified contraction for every $0 < dt \leq 1$.¹⁹*

For the hypothesis, the certificate makes the numerical state of a changing brain checkable: when its conditions hold, a returned decision carries a bound on its distance from equilibrium. This bound certifies the solve. It does not certify the accuracy of the learned body model or the safety of an action.

A.4 Learning by two equilibria

The free phase settles under the stimulus alone to s^0 . The positive nudged phase settles from s^0 with an extra drive of strength β on the output neurons toward a target: for a quadratic nudge the drive is $\beta(\text{target}_i - s_i)$, for a cross-entropy nudge $\beta(\text{target}_i - p_i)$ with p the softmax of the outputs at temperature T , one softmax per declared slot. The centred estimator settles a negative nudged phase at $-\beta$ from the same free state. With s^+ and s^- the two nudged equilibria, the contrast at synapse (i, j) and at neuron i is

$$\Delta_{ij} = \frac{1}{B} \sum_b \frac{s_{b,i}^+ s_{b,j}^+ - s_{b,i}^- s_{b,j}^-}{2\beta}, \quad \Delta_i = \frac{1}{B} \sum_b \frac{s_{b,i}^+ - s_{b,i}^-}{2\beta}. \quad (5)$$

Reciprocal synapses share one parameter and receive the mean of their two contrasts; a cap bounds every efficacy.

Lemma A.7 (locality). *There is a fixed function g of four real numbers such that the contrast of synapse (i, j) equals $g(s_i^+, s_j^+, s_i^-, s_j^-)$; the contrast is symmetric in i and j ; and if the two compared equilibria agree, every contrast is zero.²⁰*

The lemma describes the information used by the update. Reverse-mode differentiation retains or re-computes intermediate activations; the equilibrium contrast instead reads each synapse’s endpoints in the perturbed states. This can reduce storage when differentiating through many settling iterations. Both perturbed equilibria must be computed, and implicit differentiation offers another way to avoid retaining the iteration history [3].

Fact A.8 (the contrast is the gradient). *On a smooth stable equilibrium branch with symmetric effective recurrent weights, converged phases and no adaptation, the limit $\beta \rightarrow 0$ of the contrast is minus the gradient of the nudge’s loss with respect to the weight of a reciprocal pair [30]; the centred estimator cancels the first-order bias in β [16]. The library’s tests check the identity numerically: at $\beta = 10^{-4}$ and phase residuals below 10^{-11} the contrast equals the finite-difference gradient to a relative tolerance of 10^{-5} .*

¹⁸Lean: `lesion_bound, lesion_contracting`.

¹⁹Lean: `activation_lipschitz, activation_abs_le, activation_zero, learningActivation_lipschitz, learning_model_certified`.

²⁰Lean: `contrast_local, contrast_symm, contrast_eq_zero_of_agree`.

Free and nudged phases are the deterministic form of the two phases of the Boltzmann machine [1, 19], equivalent to backpropagation in a layered net [36], and the local-activation-difference rule of generalized recirculation [27]. Reward uses the three-factor form: an eligibility trace $e \leftarrow \gamma \lambda e + \Delta$ of the same contrast, a linear critic with its own trace, and one broadcast error $\delta = r + \gamma V(s') - V(s)$, with the update $\eta \delta e$ [8, 32, 33].

Intuition: let a small correction travel through the circuit. The target gently pushes the outputs, and recurrence transmits that change to the rest of the net. A synapse compares how strongly its two endpoints coactivate under the positive and negative pushes. Their difference tells it which direction the connection should move under the assumptions of Fact A.8. Credit reaches the synapse through the circuit’s changed state, so the update itself needs only its endpoints. There is a cost: the learner must solve the perturbed equilibria, and a locally computed update can be slower than a conventional backward pass. An eligibility trace carries that local comparison forward when reward arrives later; it does not guarantee that one delayed scalar identifies the right cause. The advantage is the form and storage of the update, with learning quality established by the controlled comparisons.

B Optional record algebra and capacity

A record cortex keeps, for each of its cells, what followed the readings that cell took part in. A reading $x \in \mathbb{R}^d$ of declared source neurons is made mean-free by a running mean $\bar{x}$, projected by a fixed random matrix R onto $D \gg d$ cells, rectified, and reduced to the a strongest entries at unit length,

$$k(x) = \text{kwta}_a(\max(R(x - \bar{x}), 0)). \quad (6)$$

Each predicted field holds one table $M \in \mathbb{R}^{D \times m}$. The prediction is the read $k(x)M$, and the witnessed outcome y is written by the delta rule [12, 35] at rate η ,

$$M \leftarrow M + \eta k(x)^\top (y - k(x)M). \quad (7)$$

Consequence fields use a slow rate so that a regularity is averaged over many outcomes; valued fields such as reward use rate one, so that one exposure writes. The optional record store is distinct from the neural core and from the fixed-model Gaussian boundary.

Theorem B.1 (one-shot memory and its interference law). *For any record M , keys k, q , value y and rate η over a commutative ring: (i) the read at q after the write changes by $\eta(q \cdot k)$ times the correction $y - kM$; (ii) if $k \cdot k = 1$, the read at k after a write at rate one is exactly y ; (iii) if $q \cdot k = 0$, the read at q is unchanged; (iv) if $k \cdot k = 1$, writing the same correct observation twice at rate one changes nothing.*²¹

Theorem B.2 (how far one write reaches). *Over the reals, the change a write at k makes to the read at q has sup norm exactly $|\eta| |q \cdot k|$ times the sup norm of the correction, and an overlap of at most ϵ bounds it by $|\eta| \epsilon$ times that norm.*²²

Theorem B.3 (exact storage and capacity). *After writing N pairwise orthogonal unit keys in sequence at rate one, every one of them reads its own value, whatever the initial record was; and over a field at most d keys can be pairwise orthogonal unit vectors in a d -dimensional key space.*²³

Theorem B.4 (pattern separation). *Two keys whose supports are disjoint are orthogonal, so a write at one leaves the read at the other exactly unchanged; a family of unit keys with pairwise disjoint supports, written in sequence at rate one, is stored exactly from any initial record; and a code of width D with a active cells admits at most $\lfloor D/a \rfloor$ such keys.*²⁴

²¹Lean: `read_write, read_write_self, read_write_orth, repeat_write_stable`.

²²Lean: `read_write_dist, read_write_dist_le`.

²³Lean: `read_writeUpTo, capacity_le`.

²⁴Lean: `dotProduct_eq_zero_of_disjoint_support, read_write_of_disjoint, read_writeUpTo_of_disjoint, disjoint_family_card_le`.

Intuition: a correction follows the overlap of two addresses. Suppose a reading activates two cells equally, giving a unit key $k = (1, 1, 0, 0)/\sqrt{2}$. A write at rate one corrects that reading in a single step. A second reading with key $q = (0, 0, 1, 1)/\sqrt{2}$ shares no cells and keeps its prediction exactly. A third, $q = (0, 1, 1, 0)/\sqrt{2}$, shares one cell and receives half the first reading’s correction, since $q \cdot k = 1/2$. This is the interference law in a small example: the same overlap that permits generalisation can overwrite another fact.

The four theorems explain why a useful code can learn quickly from a stream: each observation directly corrects the relevant prediction while disturbing other predictions in proportion to their key overlap. Sparsity alone guarantees neither distinct addresses nor retention; two sparse keys can be identical, and finite capacity limits exact separation. With an appropriate code, direct writes can need fewer observations than fitting the same associations in distributed weights. The record theorem does not by itself prove a replay requirement for another architecture. The published examples document measured uses of these records: a composer whose held-out accuracy is measured with the records writing online [R99], soft bodies that write records during a life and never inherit them [R100], and a Connect Four value patch whose school leaves the store empty: in three runs that wrote every schooled position into a store of 4,096 cells, the slow readout alone named the winner about three points more often on positions never seen and two points more often on positions witnessed [R101]. In the same example two positions that differ by one stone share 51 percent of their active cells, and a write to one moves the other 53 percent as far, in line with the interference law above; observing a position together with its neighbours, each anchored at its current value, moves it by 0.71 and then by 0.04 [R101].

Theorem B.5 (attention is one read of a record). *Let $M = \sum_t k_t^\top y_t$ be the additive record of N key/value pairs. The read at query q is $qM = \sum_t (q \cdot k_t) y_t$, the stored values weighted by the query/key dot products. If the keys are one-hot at distinct addresses, the read at a one-hot query is exactly the stored value, for every N up to the size of the address space.*²⁵

The first sentence is unnormalised linear attention [2, 11, 31]; the softmax read of a transformer differs by an exponential kernel and a normaliser, and modern Hopfield networks show that it is one step of an energy network [13, 14, 28]. The residual write of equation (7) is the delta rule over a sequence [37]. The identity covers additive dot-product attention, not the general softmax operation. Unlike an additive write, the residual write corrects its current prediction.

C Wiring and a declared cost model

Theorem C.1 (the certificate under evolution). *Let W satisfy Theorem A.1 with row mass ρ . (i) Every W' with $|W'_{ij}| \leq |W_{ij}|$ for all i, j satisfies it with the same constant: pruning a synapse, masking a projection, lowering a gain and scaling by any $|g| \leq 1$ are instances. (ii) If V has row mass μ and $L(\rho + \mu) < 1$, then $W + V$ settles by a certified contraction with constant $1 - \text{dt}(1 - L(\rho + \mu))$: growth by new projections stays certified inside the row-mass budget. (iii) Every member of a population whose genomes share one row-mass bound settles by one certificate, whatever its individual wiring.*²⁶

The row-mass budget certifies convergence of admitted recurrent weights. It does not establish useful specialization or certify a changing nonlinear memory basin.

C.1 Cost

Fix the declared cost model. A patch net with E synapses that settles in k steps costs kE multiply-adds per decision, plus Dm for a record read and the same for a write, and holds n potentials and Dm numbers whatever has happened before. A transformer with ℓ layers of width d over a window of T tokens costs about $\ell(12d^2 + 2Td)$ multiply-adds per token and holds a key/value cache of $2\ell Td$ numbers.

Theorem C.2 (what grows and what does not). *Under that model, the arithmetic a transformer spends on a stream of T tokens is $\ell(12d^2T + dT(T - 1))$, which grows quadratically in T , and its cache grows without bound. The arithmetic a patch net spends is $T(kE + 2Dm)$ and its state is constant in T . For every choice of positive ℓ, d, k, E, D, m there is a horizon beyond which the patch net’s total is the smaller of the two.*²⁷

²⁵Lean: `read_hebb, one_hot_recall`.

²⁶Lean: `rowMassLE_of_abs_le, rowMassLE_smul, rowMassLE_add, mutation_contracting, growth_contracting, population_contracting`.

²⁷Lean: `attention_total, attention_total_ge_quadratic, record_total, record_state_const, exists_crossover`.

The theorem is arithmetic about the declared model, not a claim that the two objects compute the same function. It states where the shapes differ: one of them re-reads its past, the other has written its past into a fixed store. Measured costs depend on the numerical solver, query accuracy and full controller.

Intuition: compression trades history for a bounded state. For a repeated body-control task, a learner may need the present position and a learned map from actions to consequences, rather than every observation that established the map. Records and traces carry that summary between decisions. Their fixed size explains the linear stream cost in the theorem, and also imposes a limit: they cannot preserve arbitrarily many independent facts exactly. The quadratic comparison assumes attention to an expanding history; a transformer with a fixed window also has linear total cost in stream length. Whether retaining the summary is better depends on what the task requires, the cost of finding the representation, the achieved error and the measured latency.

D Theorem map and the Lean library

Every Theorem, Lemma and Corollary in the paper names the Lean declaration that proves it. The public Cadence library includes the certified recurrence, activation, local contrast and wiring results, together with the representation, noise, basin, specialization and temporal-overlap boundaries [20]. The optional linear-record algebra has a separate import; it is not a component of **PatchNet**. These sources pin Lean 4.29.1 and Mathlib at the matching release. Their axiom audit allows only propositional extensionality, choice and quotient soundness, with no admissions or project-specific axioms. The historical cost results are retained in the full manuscript’s proof collection; exploratory structural-memory, readable-trace, causal-path and quadratic-boundary results have separate sources and verification in the accompanying evidence [21]. The causal-path, quadratic-boundary and protected-path lemmas are included in the public package’s 169 audited statements; other exploratory collections remain separate. The full manuscript build checks these additional sources as well. None of these collections is a formal refinement proof of the floating-point experiments or the complete library interface.

Table 4: The theorem map. Settling theorems hold for Lipschitz activations; the Activation module instantiates them for this neuron. Memory identities hold over commutative rings, the capacity bound over fields, the cost statements over the naturals.

paper	module and declaration	content
Thm A.1	Settling, <code>settle_contracting</code>	contraction with constant $1 - dt(1 - L\rho_W)$
Cor A.2	Settling, <code>equilibrium_unique</code> , <code>silence</code>	any fixed point is the equilibrium; rest is the equilibrium of no input
Thm A.3	Settling, <code>apriori_bound</code> , <code>aposteriori_bound</code>	the two error certificates
Thm A.4	Settling, <code>equilibrium_lipschitz_stimulus</code> , <code>warm_start_bound</code>	the equilibrium moves by at most $dt \delta / (1 - q)$; k warm steps leave $dt \delta q^k / (1 - q)$
Thm A.5	Settling, <code>lesion_bound</code> , <code>lesion_contracting</code>	a lesion moves the equilibrium by at most $dt \mu_S A / (1 - q)$
Thm A.6	Activation, <code>activation_lipschitz</code> , <code>activation_abs_le</code> , <code>activation_zero</code> , <code>learningActivation_lipschitz</code> , <code>learning_model_certified</code>	the neuron’s Lipschitz constant, range and rest; row mass below two certifies the learning model
Lem A.7	Credit, <code>contrast_local</code> , <code>contrast_symm</code> , <code>contrast_eq_zero_of_agree</code>	locality, symmetry, zero contrast for identical states
Thm B.1	Memory, <code>read_write</code> , <code>read_write_self</code> , <code>read_write_orth</code> , <code>repeat_write_stable</code>	interference law, one-shot write, orthogonal invariance, idempotence
Thm B.2	Sparse, <code>read_write_dist</code> , <code>read_write_dist_le</code>	the exact size of one write’s effect on another read
Thm B.3	Memory, <code>read_writeUpTo</code> , <code>capacity_le</code>	exact storage of an orthonormal family; at most d such keys

paper	module and declaration	content
Thm B.4	Sparse, dotProduct_eq_zero_of_disjoint_support, read_write_of_disjoint, read_writeUpTo_of_disjoint, disjoint_family_card_le	disjoint supports never interfere; a disjoint family is stored exactly; a code of width D with k winners holds at most $\lfloor D/k \rfloor$ of them
Thm B.5	Memory, read_hebb, one_hot_recall	the additive read is linear attention; one-hot recall is exact
Thm C.1	Wiring, rowMassLE_of_abs_le, rowMassLE_smul, rowMassLE_add, mutation_contracting, growth_contracting, population_contracting	pruning, masking and rescaling keep the certificate with the same constant; growth keeps it inside the row-mass budget; one bound certifies a whole population
Thm C.2	Cost, attention_total, attention_total_ge_quadratic, record_total, record_state_const, exists_crossover	the transformer’s arithmetic over a stream is quadratic in its length and its cache unbounded; the patch net’s is linear with constant state; a crossover exists

Three further results are used as facts and are Lean-checked in the observer-patch corpus rather than in this package [23, 25]: local repair to consensus on a federation of patches computes every finite Boolean function (`echosahedral_universality`); a work-conserving scheduler on a ranked repair system terminates without a fairness premise; and the accepted repair steps of a fixed input-independent federation number at most $n(n+1)/2$, order-sharp. No embedding of those discrete programs into this neuron model is supplied.

E Evidence map

Empirical numbers are read from the named receipts copied into the evidence package, with each receipt’s own digest where it has one, its implementation version or source hashes, and the sha256 of the copied bytes recorded in the package manifest. Quoted external results have bibliographic citations. Exact mathematical constants follow their stated definitions and hypotheses. The memory-proof sources are copied under the manifest label `mission-memory-proofs`.

The causal-admission and partitioned-phrase packets include numerical observations, checkpoints, archived implementations and verification receipts. Source hashes identify the recordings used for their audio checks; raw recordings and rendered audio are excluded from these packets. Numerical replay and reproduction of the audio from its identified source therefore have different input requirements. Deterministically compressed copies retain the original source-byte hashes in the manifest.

Four examples are published with their sources, pages and checks at <https://github.com/muellerberndt/cadence-examples>, pinned here at one commit. A worm whose brain is the measured *C. elegans* connectome takes a lesson whenever food or pain arrives and drives its body from its command interneurons; its receipt holds the browser brain and body to the library and to the reference body, and a receipt of conditioning with controls is work in progress [R98]. A composer starts from silence and computes a track, hearing each half-beat it plays [R99]. Soft bodies inherit their shape, muscles and cortices, learn during a life and plan their motor through a learned model while every unit of computation costs mass [R100]. A Connect Four brain plans by a supplied search over imagined boards that reads a value learned from a perfect player’s games [R101]. Each states what is supplied, what is learned, what was measured and what it does not show.

Table 5: Receipt identifiers and manifest labels. The library column names the release whose sources the receipt binds; OPN identifies the earlier observer-patch-nets library. The manifest preserves exact source references and copied-file hashes.

id	manifest label	receipt	library
R1	recall	associative recall over 128 pairs, two transformer arms	0.5.0
R3	pattern-separation	correlated keys under revision streams, six arms, five seeds	0.9.0
R5	digit-span	digit span, six arms, fade sweep	0.7.0
R6	n-back	n -back, four arms, two baselines	0.7.0

id	manifest label	receipt	library
R8	adaptive-depth	compute follows change: warm steps, budget, bounds, three row masses, five seeds	0.9.0
R10	event-repair	event-driven repair on the fly graph, 135 conditions	0.8.1
R12	surprise	settling steps against entropy	0.5.0
R13	matched-learning	three learning rules on one 48-neuron mechanism	0.8.1
R16	lesion	damage and recovery, three seeds	0.5.0
R17	permuted-mnist	thirty permuted tasks, six arms	0.5.0
R22	fly-counterfactuals	whole-fly counterfactual screening	0.8.1
R23	opn	fly wiring against shuffled, conformance	OPN
R24	mouse	mouse labyrinth, connectome against shuffled and a network	0.7
R26	player	self-play of the many-games brain	0.7
R27	player	sixteen games at once with a dopamine floor	0.8
R31	sizing	text width sweep, per-run files	0.4 to 0.7
R32	author	language pilots at level one	0.7
R33	hopper, pong	control gates, logs	0.5.0
R34	repeat-previous, slippery-ant	partially observed cards; slippery ant	0.7 to 0.8.1
R36	split-mnist	split MNIST, one head	0.5.0
R37	worm-probe	supplied worm mechanism under lesions	0.8.1
R41	ep-inputs	fixed-input derivatives and free/free asymmetry control	0.9.0
R42	rehearsal	split and permuted MNIST at disclosed memory and work budgets	0.9.0
R43	content-prototypes	noisy cues, fixed prototypes, aliases and capacity	0.9.0
R44	policy-credit	delayed choices, loss-policy correction and score control	0.9.0
R45	sequence-readback	causal held-out text and validation grids	0.9.0
R46	drift-diagnostic	thirty tasks and equal-budget reset forks	0.9.0
R57	instrument	the instrument: five acceptance seeds, ten gates	0.9.0
R58	composer-stage	the chip composer: five acceptance seeds, thirteen gated and seven open measures	0.9.0
R61	wiring-search	wiring search on four families and on the composer's streams: evolution, random search, hand-found wirings, intervention states	0.9.0
R62	scaling-curves	fourteen arms on four task families at four budgets, preregistered floors, retention read-back	0.9.0
R63	combined-learner	banks beside a gradient patch with a guard, ten arms, four budgets	0.9.0
R64	evolved-development	the evolution loop over wiring and development policy: contract and calibration	0.9.0
R65	composer-curves	the chip composer as curves over a life: three wirings, two sizes, two life lengths, four seeds	0.9.0
R66	player-curves	the 1942 records controller and a PPO convolutional net at matched frames, three seeds	0.9.0
R69	player-1943	the 1943 imitation player and its practice on 1942, frozen tests on eight starts	0.9.0
R71	mission-basin	nonlinear cue persistence, opposing evidence and continued-learning stress; source, full receipt and independent verification	OPN
R72	mission-specialization	nine-patch completion and warm revision; source, full receipt and independent verification	diagnostic
R73	mission-structural	normalized-contact repair, continuous controls and shared-connection failure; full receipt and verification	diagnostic
R74	mission-active-evidence	exact finite selective-inspection reference; frozen model, exhaustive enumerator and receipt	reference
R75	library-memory	reciprocal library core without temporal boundary; three-seed acquisition, retention, cue and continuation checks with archived sources	core
R76	library-temporal-boundary	same declared library tasks with held temporal overlap; finite context, cold-acquisition failures and archived sources	core
R77	amen-patch-features	fixed-seed recorded-audio feature prediction with and without temporal overlap; matched autoregressive controls, arrays and source snapshots	core
R78	library-selective-retention	absent-boundary, fixed-input and calibrated selective-plasticity tests; complete outcomes, checkpoints, sources and independent endpoint verification	core
R79	scalar-continuation	periodic continuation with current, temporal, explicit-history and supplied-clock inputs; frozen protocol, full receipt and arithmetic verification	core

id	manifest label	receipt	library
R80	readable-trace	six scalar trace-repair lemmas; pinned sources and compiled axiom audit	formal
R81	intent-control	brief-cue motor/consequence test, matched trace ablation, private branches, continued-state acquisition and linear-decoder diagnostics	core
R82	intent-episodic	independent episodic intentions with fresh-state training; trace ablation and supplied intention/position acquisition control	core
R83	temporal-repair	causal-path uniqueness proof; linear and nonlinear temporal credit, initialization comparison and fresh motif validation	diagnostic
R84	temporal-actor	linear body acquisition by centered detuning; joint action planning, actual feedback, cold restoration and independent execution audits	diagnostic
R85	temporal-observer	noisy past-state repair and one-record Gaussian-prefix equivalence; actual records, model binding, controls and independent all-history audits	diagnostic
R86	quadratic-boundary	six scalar fixed-model quadratic elimination and observation-admission theorems; compiled axiom audit	formal
R87	revision-exposure	fresh crossed static/warm exposure and gradient/Newton repair; complete memory outcomes and independent endpoint arithmetic	diagnostic
R88	amen-temporal-capacity	real-audio intention-conditioned finite paths; all-seed acquisition, retention controls and independent endpoint/recurrence checks	capacity
R89	amen-temporal-retention	local response constraints, new-phrase learning, protected-path Lean induction, sample listening and independent checks	conditional
R90	amen-sample-events	two recorded-bar recalls, supplied sample actions, known-action controls and independent neural/audio replay	capacity
R91	amen-sample-retention	sampled-action retention, protected-path conditioning, matched control and independent neural/PCM checks	bound
R92	nonlinear-input-planning	continuous input detuning, acquired nonlinear-body control, full training/plan replay and actual readback controls	conditional
R93	amen-sample-repertoire	joint four-bar recorded repertoire, all taught sample identities, untaught cue interpolation and exact audio replay	capacity
R94	amen-sample-metric	matched constrained-readout metric, complete update and PCM replay, conditioning and finite-detuning counterexample	conditional
R95	causal-admission	observed-loss parameter admission, acquired acoustic control and fixed-reference restoration; bounded outcomes and independent replay	conditional
R96	partitioned-cue	masked joint recurrence, capped dense comparison, complete update replay and untrained radius-normalization controls	diagnostic
R97	self-readback-control	exact matrix readback equivalence and finite temporal sensitivity controls	diagnostic
R98	example-worm	published example: the <i>C. elegans</i> connectome as one masked temporal patch net; browser brain and body held to the library and to the reference body	main
R99	example-amen	published example: a composer in one record patch; next-event accuracy on held-out tracks against baselines, browser engine held to the library	main
R100	example-patchworld	published example: soft bodies with two record patches under selection; conserved mass, the founders' model on a held-out stream, two worlds of 20,000 ticks	main
R101	example-connect4	published example: a value patch schooled by a perfect player and a supplied search; 200 games with every move graded by the solver, the search alone as control, the record store under bulk writing, and what neighbouring positions share in it	main

F Notation

n neuron count; v_i potential; $s_i = \text{act}(v_i)$ activation; W_{ij} effective weight from j to i ; c_i stimulus including bias; dt step; ρ_W maximum absolute incoming weight sum; L activation Lipschitz constant; q contraction constant; v^* equilibrium; β nudge strength; $s^\pm$ nudged equilibria; Δ contrast; η learning rate; M record table; k, q keys; y value; D code width; a active code cells; $k(x)$ reading code; R code expansion; e eligibility trace; δ reward error or stimulus change as stated; γ, λ discount and trace decay; μ added-projection row mass; ℓ, d, T transformer layers, width and window in the cost model.